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C++ Programming
Scope lifetime and void functions

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What does mean by variable Scope?

- The boundary or area in which variables are accessible after the declaration are the scope of these variables.
- Variable scope determine the accessibility of variables
- There are two kinds of variable with respect to scope
 1. **Local Variable**
 2. **Global Variable**

Local Variables

- The variable which are declared inside the block or function body is called local variables.

```
1  #include <iostream>
2  using namespace std;
3  int main ()
4  {
5      // Local variable declaration:
6      int n1, n2;
7      n1 = 20;
8      n2 = 30;
9      {
10         // one more variable with different scope
11         int n3=100;
12         cout<<n3<<endl;
13     }
14     cout<<n1 + n2;
15     return 0;
16 }
```

Global Variables

- The variable which are declared outside from all the functions and accessible throughout the program are known as global variables

```
#include <iostream>
using namespace std;
// Global variable declaration:
int num1;
int main ()
{
    // Local variable declaration:
    int num2;
    num1 = 20;
    num2 = 30; |
    cout<<num1 + num2;
    return 0;
}
```

What does mean Life time of variables

- Life time of variable mean duration for which variable exist in memory
- Variable life time start when a memory is allocated to a variable
- Variable life time ends when a memory is deallocated from variable

Life time of local variable

- Life time of local variable start when control enter inside the function or block where variable is declared
- Life time of local variable ends when control exit from the function or block where variable is declared

Lifetime of Global variable

- Lifetime of global variable start when program execution start
- Lifetime of global variable ends when program execution end

What is void in C++?

- In C++, void is a special data type that means "nothing" or "no type".
- It is used in two main ways:
 - ✓ As a function return type (function returns nothing)
 - ✓ As a function parameter list (function takes no parameters)

void as a Return Type

- When a function is declared with void as its return type, it means the function performs an action but does not return a value.

```
#include <iostream>

void greetUser() {
    std::cout << "Welcome to the program!" << std::endl;
}
```

void in the Parameter List

This is used to say a function takes no arguments (no inputs). In C++, it's optional to use void in the parentheses; writing empty () means the same thing.

•

```
void showCredits(void) {  
    std::cout << "Welcome to the program!" << std::endl;  
}
```

void in the Parameter List Cont..

void: means the function doesn't return anything.

showCredits: the name of the function.

(void): means it takes no parameters (C-style syntax).

Inside the function, it prints a welcome message using `std::cout`.

Thanks ?